

SEQUENTIAL PROCEDURES WITH ELIMINATION FOR SELECTING THE BEST OF k NORMAL POPULATIONS

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SUMMARY

Sequential procedures for selecting the normal population with the largest mean among k normal populations are studied. Various assumptions are made on the variances of the population, including the possibility that they are unknown and unequal. The procedures are based on the concept of a *selection sequence*, i.e. a sequence of subsets of the k populations with the property that the best population is contained in every member of the sequence with given probability for all parameter points outside an indifference zone. The procedures provide a subset of given size (which may be taken to be one) which contains the best population with given probability. Asymptotic expressions are obtained for the expected sample sizes of the procedures which show that, asymptotically, our procedures are more efficient than existing procedures. Selection sequences are also constructed such that no indifference-zone assumption is needed.

1. Introduction

In this article we consider the problem of selecting the "best" of $k \geq 2$ normal populations, that is, the population with the largest mean. Various formulations of this problem have received much attention in the statistical literature. The book of Bechhofer, Kiefer and Sobel (1968) contains references to most of the available theory.

The problem has been formulated in basically two ways, namely the "indifference zone" formulation, due to Bechhofer (1954) and the *subset formulation* due to Gupta (1965). Both these formulations and the proposed solutions have some unpleasant features. A solution to the indifference zone formulation guarantees a minimum probability of a correct selection for all parameter points not in an indifference zone. However, the specification of this zone may be difficult in applications. On the other hand, a solution to the subset formulation provides a subset of the populations which contains, with given probability, the best population. However, the size of the subset is random and may be large.

We shall propose various procedures using the concept of a *selection sequence*. (cf. Swanepoel and Geertsema (1973).) Let $\pi_1, \dots, \pi_k$ be normal populations, such that π_i is $N(\mu_i, \sigma_i^2)$, $i = 1, \dots, k$. We assume, without loss of generality that $\mu_1 \leq \dots \leq \mu_k$. The parameters $\mu_1, \dots, \mu_k$ are unknown and various assumptions will be made about $\sigma_1^2, \dots, \sigma_k^2$,

including a case where the σ 's will be assumed unknown and possibly unequal. We refer to π_k as the best population. (We realise that in the case where the variances are possibly different, there is some question about what should be meant by a "best" population. However, in practice, the variances are probably never exactly equal and in a case where the variances do not differ too much our definition of the "best" population should still be realistic and our procedure applicable.)

Let $\{\Pi_n\}$ be a sequence of subsets of $\pi_1, \dots, \pi_k$ such that $P(\pi_k \in \Pi_n \text{ for all } n \geq 1) \geq P^*$ for $\theta \in \Theta_0$ where $\theta = (\mu_1, \dots, \mu_k, \sigma_1^2, \dots, \sigma_k^2)$ and Θ_0 is a subset of the parameter space Θ . We shall refer to such a sequence as a selection sequence for $\theta \in \Theta_0$. Typically Π_n will be based on at most n observations from each population. The concept of a selection sequence is closely related to that of a *confidence sequence* introduced by Robbins (1970) in connection with confidence intervals. The advantages of a selection sequence are then similar to those of a confidence sequence. It allows one to follow the unknown π_k with a subset Π_n which typically shrinks to a subset with one element in such a way that the probability is at least P^* that Π_n contains π_k at each stage. It is then also true that $P(\pi_k \in \Pi_N) \geq P^*$ for $\theta \in \Theta_0$ even if N is a random variable that depends on the observations. In the construction of selection sequences in this paper, we incorporate the desirable feature of elimination of inferior populations as the sampling proceeds. The idea of elimination is due to Paulson (1964) and we remark that in his paper a selection sequence is used implicitly.

In cases where the selection sequence eventually shrinks to one element only, we will define a sequential procedure as follows: Let N be the first integer n such that Π_n contains at most m elements ($1 \leq m \leq k-1$). Then Π_N is a subset containing the best population with probability at least P^* for $\theta \in \Theta_0$ and Π_N consists of at most m elements. If we take $m = 1$, we obtain a sequential selection procedure with elimination which chooses the best population with given probability of a correct selection for $\theta \in \Theta_0$. In particular if $\sigma_1^2 = \dots = \sigma_k^2 = \sigma^2$ with σ^2 known, we obtain a competitor for Paulson's procedure and an asymptotic analysis shows that our procedure is asymptotically superior as $P^* \rightarrow 1$, where P^* is the required probability of a correct selection. The only available procedure for the case σ^2 unknown is due to Robbins, Sobel and Starr (1968). However, our procedure of section 2 improves on this procedure in several ways and is also asymptotically superior as $P^* \rightarrow 1$. Dudewicz and Dalal (1971) claim in an abstract that they have a two-stage procedure for the case where the σ 's are not necessarily equal. However, it is to be expected that a fully sequential procedure should be more efficient than a two-stage procedure.

Finally, we remark that selection sequences will be constructed which satisfy $P(\pi_k \in \Pi_n \text{ for all } n \geq 1) \geq P^*$ for *all* parameter points. Use of such a sequence does not require the specification of an indifference zone and the computation of the sequence

step by step is in a sense an improvement of Gupta's procedure, since it will be shown that $\{\Pi_n\}$ shrinks except in a few exceptional cases. Furthermore the sampling and computation process may be stopped at any arbitrary point (which may depend on the observations) and the resulting subset contains π_k with probability at least P^* .

Tables for use in practical applications and some Monte Carlo results are given at the end of the paper.

2. A Selection Sequence for the General Case

In this section we assume that $\mu_1, \dots, \mu_k, \sigma_1^2, \dots, \sigma_k^2$ are all unknown.

We define a selection sequence as follows:

Let (i) $h(t, n) = [(tn)^{n-1} - 1]^{1/2}$ for $n = 1, 2, 3, \dots$ where

$t = (1 + a^2)/2$ and $a = a(k, P^*)$ satisfies the following equation

$$\frac{1}{2} - (\arctan a)/\pi + a/(1 + a^2)\pi = (1 - P^*)/2(k - 1).$$

Clearly $h(t, n) \rightarrow 0$ as $n \rightarrow \infty$ for fixed t and $h(t, n) \uparrow \infty$ as $P^* \uparrow 1$ for n fixed.

Also $\pi a(1 - P^*)/4(k - 1) \rightarrow 1$ as $P^* \rightarrow 1$.

$$(ii) \bar{X}_i(n) = n^{-1} \sum_{j=1}^n X_{ij}, \quad i = 1, \dots, k$$

$$(iii) H(j, i, n) = [n^{-1} \sum_{\beta=1}^n \{(X_{j\beta} - X_{i\beta}) - (\bar{X}_j(n) - \bar{X}_i(n))\}^2]^{1/2} \quad (j \neq i)$$

where $X_{\alpha\gamma}$ is the γ -th observation from population π_α ($\alpha = 1, \dots, k$).

We start by taking one observation from each population and let Π_1 be the set of all populations $\pi_1, \dots, \pi_k$. Let now $n \geq 2$ and at stage n take one observation from each population in Π_{n-1} where we define recursively

$$(1) \quad \Pi_n = \{\pi_s : \bar{X}_s(n) > \bar{X}_1(n) - h(t, n)H(i, s, n) + \Delta^*, s \neq i, s \in I_{n-1}\} \cup \{\pi_i\}$$

where $\bar{X}_i(n) = \max_{j \in I_{n-1}} \bar{X}_j(n)$, where $\Delta^* > 0$, and where

$$I_n = \{s : \pi_s \in \Pi_n\}, I_1 = \{1, \dots, k\}.$$

If, at some stage n , $\pi_j \in \Pi_{n-1}$ but $\pi_j \notin \Pi_n$ (for some $j = 1, \dots, k$) we say that π_j is *eliminated* at stage n .

In order to prove that $\{\Pi_n\}$ is a selection sequence we need the following result due to Robbins [(1970), p. 1402].

Lemma 1.

Let $\{X_n\}$ be a sequence of independent random variables, each having a $N(\mu, \sigma^2)$ distribution. Let $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$ and $v_n^2 = n^{-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$, then

$$P(|\bar{X}_n - \mu| \geq v_n [(dn)^{n-1} - 1]^{1/2} \text{ for some } n \geq 2) \\ \leq 1 - (2 \arctan x)/\pi + 2x/\pi(1 + x^2) \quad (x > 0)$$

$$\text{where } d = \frac{1}{2}(1 + x^2)^2.$$

Theorem 1

The sequence $\{\Pi_n\}$ defined in (1) is a selection sequence for $\mu_k - \mu_{k-1} \geq \Delta^*$.

Proof:

$$P(\pi_k \notin \Pi_n \text{ for some } n \geq 1)$$

$$\leq P(\text{For at least one value of } s (s = 1, \dots, k-1))$$

$$\bar{X}_k(n) \leq \bar{X}_s(n) - h(t, n)H(s, k, n) + \Delta^* \text{ for some } n \geq 2)$$

$$\leq \sum_{s=1}^{k-1} P(\bar{X}_k(n) \leq \bar{X}_s(n) - h(t, n)H(s, k, n) + \Delta^* \text{ for some } n \geq 2)$$

$$\leq \sum_{s=1}^{k-1} P(\bar{X}_s(n) - \bar{X}_k(n) - (\mu_s - \mu_k) \geq h(t, n)H(s, k, n) \text{ for some } n \geq 2)$$

$$\leq (k-1)(1-P^*)/(k-1)$$

$$= 1 - P^*$$

using (i) and lemma 1.

A sequential procedure based on the selection sequence

Let $N = N(\Delta^*, P^*)$ be the first $n \geq 2$ such that $S_n \leq m$ where S_n is the number of populations in Π_n and where m is an integer chosen in advance, $1 \leq m \leq k-1$. Choose the subset Π_N of populations as subset which contains the best population.

We observe that the subset of populations obtained has at most m elements and it follows immediately from the defining property of the selection sequence $\{\Pi_n\}$ that if CS denotes a "correct selection" (i.e. that $\pi_k \in \Pi_N$) that $P(\text{CS}) \geq P^*$ for $\mu_k - \mu_{k-1} \geq \Delta^*$. Note also that we may take $m = 1$, thus obtaining a procedure which selects a single population as the best.

In order to prove that the sequential procedure terminates with probability one and for other purposes as well, we define the following auxiliary stopping variable: Consider the case where at each stage of sampling an independent observation is obtained from *each* of the k populations and let M be the first $n \geq 2$ such that

$$(2) \quad h(t, n) \sum_{i=1}^k \sum_{j=1}^k H(j, i, n) < \Delta^*.$$

Lemma 2

$N \leq M$ for all $\theta \in \Theta$.

Proof

It follows from (2) that

$$h(t, M) \sum_{i=1}^k \sum_{j=1}^k H(j, i, M) < \Delta^*.$$

Hence $h(t, M)H(j, i, M) < \Delta^*$ for all (i, j) , $i, j = 1, \dots, k$. If $N \geq M$, then I_{M-1} contains at least $m+1$ indices, say $j_1 < j_2 < \dots < j_\ell$ (some $\ell = m+1, \dots, k$).

Then

$$\max_{j \in I_{M-1}} \bar{X}_j(M) \geq \bar{X}_{j_i}(M) > \bar{X}_{j_i}(M) + h(t, M)H(j_s, j_i, M) - \Delta^* \text{ for all } i = 1, \dots, \ell; s = 1, \dots, \ell$$

so that $N = M$.

Theorem 2

$P(N < \infty) = 1$ for all $\theta \in \Theta$ and for any $1 \leq m \leq k-1$.

Proof

In view of lemma 2 it is sufficient to prove that $P(M < \infty) = 1$ for all $\theta \in \Theta$ and $1 \leq m \leq k-1$.

But $h(t, n) \rightarrow 0$ as $n \rightarrow \infty$ (for t fixed) and $H(j, i, n) \rightarrow (\sigma_j^2 + \sigma_i^2)^{1/2}$ a.s. as $n \rightarrow \infty$ for all (i, j) , $i \neq j$, by the Strong Law of Large Numbers. The result follows since $\Delta^* > 0$.

Asymptotic expected number of observations for the sequential procedure

Let T denote the total number of observations taken from all populations up to stage N . We shall now develop asymptotic theory as $P^* \rightarrow 1$ which should be of importance since in practice values of P^* close to one are usually chosen. (See for instance Bechhofer, Kiefer and Sobel (1968), Ch. 6, where extensive asymptotic theory as $P^* \rightarrow 1$ is developed.) Our purpose is to obtain asymptotic expressions for EN and ET . For this purpose we need the following lemma which gives asymptotic properties of the auxiliary stopping variable M which was defined in (2):

Lemma 3

(i) $M \uparrow \infty$ a.s. as $P^* \uparrow 1$ and $EM \uparrow \infty$ as $P^* \uparrow 1$.

(ii) $M/\log t \rightarrow L$ a.s. as $P^* \rightarrow 1$ where

$$L = 1/\log(1 + (\Delta^*)^2 / (\sum_{i \neq j} (\sigma_i^2 + \sigma_j^2)^{1/2})^2).$$

(iii) The set $\{M/\log t\}$ for $P^* < 1$ is uniformly integrable.

Proof:

(i) This follows from the definition of M , the fact that $h(t, n) \uparrow \infty$ as $P^* \uparrow 1$ for n fixed and the Monotone Convergence Theorem.

(ii) From the definition of M we have $h(t, M) \sum_{i \neq j} H(j, i, M) < \Delta^*$

$$\text{i.e. } (\log t + \log M)/M < \log(1 + (\Delta^*)^2 / (\sum_{i \neq j} H(j, i, M))^2)$$

so that

$$\liminf_{P^* \rightarrow 1} M/\log t \geq L \text{ a.s.}$$

since $H(j, i, M) \rightarrow (\sigma_j^2 + \sigma_i^2)^{1/2}$ a.s. as $P^* \rightarrow 1$ for all (i, j) with $i \neq j$

since $M \rightarrow \infty$ a.s.

Using

$$h(t, M-1) \sum_{i \neq j} H(j, i, M-1) \geq \Delta^*$$

we obtain similarly

$$\limsup_{P^* \rightarrow 1} M/\log t \leq L \text{ a.s.}$$

(iii) Using a result of Bickel and Yahav (1968), it is sufficient to prove that

$$\sum_{n=1}^{\infty} \sup_{P_0^* < P^* < 1} P(M/\log t > n) < \infty \text{ for some } P_0^* < 1.$$

Let $r = r_n(t)$ be the largest integer less than or equal to $n \log t$. Then

$$\begin{aligned} & P(M > r) \\ & \leq P(h(t, r) \sum_{i \neq j} H(j, i, r) \geq \Delta^*) \\ & \leq P(h(t, r) H(j, i, r) \geq \Delta^*/k(k-1) \text{ for at least one pair } (i, j), i \neq j) \\ & \leq \sum_{i \neq j} P(h(t, r) H(j, i, r) \geq \Delta^*/k(k-1)) \\ & \leq (h(t, r) k(k-1)/\Delta^*)^4 \sum_{i \neq j} E(H(j, i, r))^4, \text{ using Markov's inequality.} \\ & = O((n^{-1} \log n)^2) \text{ uniformly in } P^* \end{aligned}$$

since $E(H(j, i, r))^4 = O(1)$ and

$$r[h(t, r)]^2/\log(tr) \rightarrow 1 \text{ uniformly in}$$

$P^* \in (P_0^*, 1)$ as $n \rightarrow \infty$ for some $P_0^* < 1$.

Theorem 3

If $\mu_k > \mu_{k-1}$, then

$$\lim_{P^* \rightarrow 1} ET/(-4 \log(1 - P^*)) \approx \sum_{i=1}^{k-m-1} y_{(i)} + (m+1)y_{(k-m)}$$

$$\lim_{P^* \rightarrow 1} EN/(-4 \log(1 - P^*)) = y_{(k-m)}$$

where $y_{(1)} \leq y_{(2)} \leq \dots \leq y_{(k-1)}$ are the values of $y_1, \dots, y_{k-1}$ arranged in increasing order, where

$$y_i = 1/\log \{1 + (\mu_k - \mu_i + \Delta^*)^2/(\sigma_k^2 + \sigma_i^2)\} \text{ for } i = 1, \dots, k-1.$$

Proof:

The method used is an extension of the method of Perng (1969). Let (Ω, Γ, P) be the underlying probability space and let

$$\Omega_0 = \{\omega \in \Omega : \pi_k \in \Pi_N\}.$$

Define on Ω_0 and for each $i = 1, \dots, k-1$ the random variable N_i to be the first integer $n \geq 2$ such that

$$(3) \quad \bar{X}_i(n) \leq \bar{X}_j(n) - h(t, n)H(j, i, n) + \Delta^* \quad (i \neq j)$$

$$\text{where } \bar{X}_j(n) = \max_{s \in I_{n-1}} \bar{X}_s(n).$$

Then N_i is the number of observations from π_j needed to eliminate π_i .

Let $N_{(1)} \leq N_{(2)} \leq \dots \leq N_{(k-1)}$ be the order statistics of $N_1, \dots, N_k$. Then

$$(4) \quad T = N_{(1)} + \dots + N_{(k-m-1)} + (m+1)N_{(k-m)} \text{ on } \Omega_0$$

From (ii) of lemma 3 and the fact that (see theorem 1) $I(\Omega_0^c) \rightarrow 0$ in probability as $P^* \rightarrow 1$ where $I(A)$ denotes the indicator function of event A , it follows that $MI(\Omega_0^c)/\log t \rightarrow 0$ in probability as $P^* \rightarrow 1$. Using (iii) of lemma 3 and the definition of uniform integrability, we obtain that the set $\{MI(\Omega_0^c)/\log t\}$ for $P^* < 1$ is uniformly integrable. Then (see Loève (1963), p. 163).

$$\begin{aligned} 0 &\leq \limsup_{P^* \rightarrow 1} \int_{\Omega_0^c} T/\log t \, dP \\ &\leq \limsup_{P^* \rightarrow 1} \int_{\Omega_0^c} (kM)/\log t \, dP \quad (\text{lemma 2}) \\ &= 0. \end{aligned}$$

Hence

$$(5) \quad \lim_{P^* \rightarrow 1} ET/\log t = \lim_{P^* \rightarrow 1} \int_{\Omega_0} T/\log t \, dP$$

provided the right-hand limit exists. We shall now establish the existence of this limit and at the same time find an expression for it.

Let i be fixed ($i = 1, \dots, k-1$) and let j be between $1, \dots, k$. From the Strong Law of Large Numbers we have

$$(6) \quad \bar{X}_j(n) \rightarrow \mu_j \text{ a.s. and } \bar{X}_i(n) \rightarrow \mu_i \text{ a.s. as } n \rightarrow \infty$$

$$(7) \quad H(k, i, n) \rightarrow (\sigma_k^2 + \sigma_i^2)^{1/2} \text{ a.s. as } n \rightarrow \infty.$$

From (6) and (7) and Egoroff's Theorem it follows that for arbitrary $\delta > 0$ there are sets A_{ji} and B_{ji} such that $B_{ji} = A_{ji}^c$, $P(B_{ji}) < \delta$ and that on A_{ji}

$$(8) \quad \bar{X}_j(n) \rightarrow \mu_j, \bar{X}_i(n) \rightarrow \mu_i, H(k, i, n) \rightarrow (\sigma_k^2 + \sigma_i^2)^{1/2} \text{ uniformly as } n \rightarrow \infty \text{ and thus also}$$

$$(9) \quad \bar{X}_j(n) - \bar{X}_i(n) \rightarrow \mu_j - \mu_i \text{ uniformly on } A_{ji} \text{ as } n \rightarrow \infty.$$

From (8) and (9) follows that there exist positive integers $n_{ji}(\delta)$ (independent of $\omega \in A_{ji}$) such that for all $\omega \in A_{ji}$ we have that for $n > n_{ji}(\delta)$

$$(10) \quad \mu_j - \mu_i - \delta < \bar{X}_j(n) - \bar{X}_i(n) < \mu_j - \mu_i + \delta$$

$$(11) \quad (\sigma_k^2 + \sigma_i^2)^{1/2} - \delta < H(k, i, n) < (\sigma_k^2 + \sigma_i^2)^{1/2} + \delta.$$

Let now $m_i(\delta) = \max_j n_{ji}(\delta)$ and

$$(12) \quad B^{(i)} = \bigcup_{j=1}^k B_{ji}, \Omega_2^{(i)} = \Omega_0 - B^{(i)}.$$

Clearly $m_i(\delta)$ and $B^{(i)}$ do not depend on P^* . From the fact that $P(B_{ji}) < \delta$ we obtain

$$(13) \quad P(B^{(i)}) \leq k\delta.$$

We thus find that (10) and (11) now hold for all $\omega \in \Omega_2^{(i)}$ and for $n > m_i(\delta)$ so that

$$\begin{aligned} & (\mu_j - \mu_i - \delta) - H(t, n)[(\sigma_k^2 + \sigma_i^2)^{1/2} + \delta] \\ & < \bar{X}_j(n) - \bar{X}_i(n) - h(t, n)H(k, i, n) < (\mu_j - \mu_i + \delta) - h(t, n)[(\sigma_k^2 + \sigma_i^2)^{1/2} - \delta] \end{aligned}$$

from which follows

$$\begin{aligned} (14) \quad & \max_{j \in I_{n-1}} (\mu_j - \mu_i - \delta) - h(t, n)[(\sigma_k^2 + \sigma_i^2)^{1/2} + \delta] \\ & < \max_{j \in I_{n-1}} \bar{X}_j(n) - \bar{X}_i(n) - h(t, n)H(k, i, n) \end{aligned}$$

$$< \max_{j \in I_{n-1}} (\mu_j - \mu_1 + \delta) - h(t, n)[(\sigma_k^2 + \sigma_1^2)^{1/2} - \delta].$$

From the fact that (8) holds on $\Omega_2^{(i)}$ for $j=1, \dots, k$ and $\mu_k > \mu_{k-1}$, we find that there is an integer $m'_1(\delta)$ (independent of $\omega \in \Omega_2^{(i)}$) such that for $n \geq m'_1(\delta)$ we have

$$\bar{X}_k(n) > \bar{X}_j(n), j=1, \dots, k-1$$

and since on $\Omega_2^{(i)}$ π_k is not eliminated, we obtain for $n \geq n_1(\delta) = \max(m_1(\delta), m'_1(\delta))$

$$(15) \quad \max_{j \in I_{n-1}} \bar{X}_j(n) = \bar{X}_k(n)$$

and also (using (14) and (15))

$$(16) \quad (\mu_k - \mu_1 - \delta) - h(t, n)[(\sigma_k^2 + \sigma_1^2)^{1/2} + \delta] < \bar{X}_k(n) - \bar{X}_1(n) - h(t, n)H(k, i, n) \\ < (\mu_k - \mu_1 + \delta) - h(t, n)[(\sigma_k^2 + \sigma_1^2)^{1/2} - \delta]$$

for all $\omega \in \Omega_2^{(i)}$.

Using (16) and the definition of N_i , we obtain on $\Omega_2^{(i)}$ either

$$(17) \quad N_i \leq n_1(\delta) + 1 \text{ or}$$

$$(18) \quad (\log t + \log N_i)x_i(-\delta) \leq N_i \leq (\log t + \log(N_i - 1))x_i(\delta) + 1$$

where $x_i(\delta) = 1/\log \{1 + (\mu_k - \mu_1 - \delta + \Delta^*)^2/((\sigma_k^2 + \sigma_1^2)^{1/2} + \delta)^2\}$.

Let now i vary from $1, \dots, k-1$ and define

$$(19) \quad B = \bigcup_{i=1}^{k-1} B^{(i)}, \Omega_2 = \Omega_0 - B, n_0(\delta) = \max_{1 \leq i \leq k-1} n_1(\delta)$$

Then, from (17) and (18) we get either

$$(20) \quad N_i \leq n_0(\delta) + 1 \text{ for some } i \text{ or}$$

$$(21) \quad (\log t + \log N_i)x_i(-\delta) \leq N_i \leq (\log t + \log(N_i - 1))x_i(\delta) + 1$$

for all $i=1, \dots, k-1$.

If (21) holds we obtain from $1 \leq N_i \leq M$ that

$$(22) \quad \log t x_i(-\delta) \leq N_i \leq (\log t + \log M)x_i(\delta) + 1 \quad i=1, \dots, k-1$$

and from (22) we get

$$(23) \quad \log t d_{(i)}(\delta) \leq N_{(i)} \leq (\log t + \log M)x_{(i)}(\delta) + 1 \text{ on } \Omega_2, i=1, \dots, k-1.$$

where the $x_{(i)}(\delta)$ are the $x_i(\delta)$ arranged in increasing order and the $d_{(i)}(\delta)$ are the $x_i(-\delta)$ arranged in increasing order.

Let

$$(24) \quad \Omega_3 = \{ \omega \in \Omega_2 : N_i(\omega) \leq n_o(\delta) + 1 \text{ for some } i=1, \dots, k-1 \}.$$

Then

$$(25) \quad \int_{\Omega_o} N_{(i)} dP = \int_{\Omega_3} N_{(i)} dP + \int_{\Omega_2 - \Omega_3} N_{(i)} dP + \int_B N_{(i)} dP$$

and

$$(26) \quad P(B) \leq k(k-1)\delta \quad (\text{from (13) and (19)}).$$

Then (25) and (23) lead to

$$(27) \quad \int_{\Omega_o} N_{(i)} dP \geq \log t d_{(i)}(\delta) P(\Omega_2 - \Omega_3)$$

$$\text{so that } \liminf_{P^* \rightarrow 1} \int_{\Omega_o} N_{(i)} / \log t dP \geq d_{(i)}(\delta)(1 - P(B)).$$

By (26) and since δ is arbitrary, we obtain

$$(28) \quad \liminf_{P^* \rightarrow 1} \int_{\Omega_o} N_{(i)} / \log t dP \geq y_{(i)}, i=1, \dots, k-1.$$

On the other hand, we also obtain from (25) and (23)

$$(29) \quad \int_{\Omega_o} N_{(i)} dP \leq \int_{\Omega_3} M dP + x_{(i)}(\delta) \int_{\Omega} (\log t + \log M) dP + 1 + \int_B M dP$$

Using (iii) of lemma 3 we have that the sets $\{\log M / \log t\}$ and $\{MI(\Omega_3) / \log t\}$ for $P^* < 1$

are uniformly integrable. Since $l(\Omega_3) \rightarrow 0$ in probability and $\log M/\log t \rightarrow 0$ a.s. as $P^* \rightarrow 1$ (using (ii) of lemma 3) we obtain

$$(30) \quad \limsup_{P^* \rightarrow 1} \int_{\Omega_0} N_{(i)}/\log t \, dP \leq x_{(i)}(\delta) + LP(B)$$

where L is defined in (ii) of lemma 3.

Since δ is arbitrary it follows from (30) and (26) that

$$(31) \quad \limsup_{P^* \rightarrow 1} \int_{\Omega_0} N_{(i)}/\log t \, dP \leq y_{(i)}, i=1, \dots, k-1.$$

The first part of the theorem now follows from (4), (5), (28) and (31) and the asymptotic expression for the function a (beginning of section 2).

Also

$$\begin{aligned} EN/\log t &= \int_{\Omega_0} N/\log t \, dP + \int_{\Omega_0^c} N/\log t \, dP \\ &= \int_{\Omega_0} N_{(k-m)}/\log t \, dP + \int_{\Omega_0^c} N/\log t \, dP. \end{aligned}$$

In the same way as (5) was obtained, we find

$$\int_{\Omega_0^c} N/\log t \, dP \rightarrow 0 \text{ as } P^* \rightarrow 1$$

so that the second part of the theorem follows from (28) and (31).

Remarks

Using theorem 3 above and theorems 3 and 4 of Swanepoel and Geertsema (1973) it is seen immediately that the asymptotic number of observations of the present procedure is never larger than the asymptotic numbers of observations of the procedures considered there.

Another procedure with which our procedure with $m=1$ may be compared, is one proposed by Robbins, Sobel and Starr (1968). We shall denote the number of observations needed by this procedure by $T_R = T_R(\Delta^*, P^*)$. This is the only procedure available which does not require knowledge of an assumed common variance σ^2 . However, our procedure does not even assume a common variance and our procedure has the desirable elimination property.

We now consider the relative efficiency of these two procedures as $P^* \rightarrow 1$ and in the least favourable configuration where $\mu_1 = \dots = \mu_{k-1} = \mu_k - \Delta^*$. We point out that the asymptotic expected sample size of our procedure has its maximum (outside the indifference zone) in this configuration, while the asymptotic expression for ET_R does not

depend on the specific configuration. In fact, it may be shown that

$$\lim_{P^* \rightarrow 1} ET_R / (-4 \log(1 - P^*)) = k\sigma^2 / (\Delta^*)^2$$

so that, in the least favourable configuration

$$\lim_{P^* \rightarrow 1} ET / ET_R = (\Delta^* / \sigma)^2 / \log(1 + 2(\Delta^* / \sigma)^2)$$

which is ≤ 1 at least for $\Delta^* \leq \sigma$.

A further advantage of our procedure is the fact that it guarantees $P(\text{CS}) \geq P^*$ for $\mu_k - \mu_{k-1} \geq \Delta^*$, while the procedure of Robbins, Sobel and Starr satisfies the requirement only *asymptotically* as $\Delta^* \rightarrow 0$. Moreover,

$$\lim_{\Delta^* \rightarrow 0} T_R = \infty \quad \text{a.s.}$$

for all parameter configurations, which is not the case with our procedure. In fact, let N' denote the stopping variable N with Δ^* replaced by 0 everywhere in its definition. Then

$$\limsup_{\Delta^* \rightarrow 0} N \leq N' \text{ for all } \theta \in \Theta,$$

and (a) $N' < \infty$ a.s. if $\mu_1 < \dots < \mu_{k-m+1} \leq \mu_{k-m+2} \leq \dots \leq \mu_k$

(this will be proved in lemma 4).

(b) $P(N' < \infty) > 0$ for other configurations.

It follows that for Δ^* small enough we have $T \leq T_R$ a.s. for typical configurations (such as those in (a)).

3. A selection sequence for the general case with no indifference zone assumption

If we retrace the steps of the proof of theorem 1 in the preceding section, we see that we may put $\Delta^* = 0$ everywhere in the proof. This implies that the selection sequence defined in (1) with $\Delta^* = 0$ remains a selection sequence which now satisfies $P(\pi_k \in \Pi_n \text{ for all } n \geq 1) \geq P^*$ for all $\theta \in \Theta$. This means that there is no indifference zone assumption.

We remark that if $\tilde{\Pi}_n$ denotes the subset of populations after n observations from each population for Gupta's procedure in the case of a common unknown variance (see Gupta (1965)) then

$$\inf_{\theta \in \Theta} P(\pi_k \in \tilde{\Pi}_n \text{ for all } n \geq 2) = 0.$$

This follows upon considering a configuration where $\mu_k = \mu_{k-1}$ and applying the Law of the Iterated Logarithm. Hence the sequence $\{\tilde{\pi}_n\}$ does not form a selection sequence for $\theta \in \Theta$.

We could attempt to base a sequential procedure on our selection sequence introduced above in the same way as was done in section 2. However, if N' is the stopping variable thus obtained, it is clear that N' need not be finite at least for configurations such that $m \leq \ell - 1$ where ℓ is the number of $\mu_i = \mu_k, i=1, \dots, k$. On the other hand, if $m \geq \ell$ and $\mu_1 < \mu_2 < \dots < \mu_{k-m+1}$ we prove in lemma 4 that $N' < \infty$ a.s. In typical cases (i.e. where $m \geq \ell$ and $\mu_1 < \dots < \mu_{k-m+1}$) one could thus proceed sampling until a subset of given size remains. In any case, the sampling may be terminated at a convenient point (which may depend on the observations) and the resulting subset contains the best population with probability at least P^* .

We next show how an asymptotic expression for EN' may be derived for those parameter configurations where

$$(32) \quad \mu_1 < \mu_2 < \dots < \mu_{k-m+1} \leq \dots \leq \mu_k,$$

and we work under assumption (32) for the remainder of this section. The method of analysis is exactly the same as that of section 2, except that a different auxiliary variable M' is used, where M' is defined as follows: Consider the case where at each stage of sampling an observation is obtained from each population, and let

$$M' = \max(R_1, \dots, R_{k-m+1}, R^*, a_0)$$

where (i) R_i is the first integer n_0 such that $|\bar{X}_i(n) - \mu_i| < \delta$ for all $n \geq n_0$ ($i=1, \dots, k-m+1$) with

$$\delta = \min_{2 \leq i \leq k-m+1} (\mu_i - \mu_{i-1})/3.$$

(ii) R^* is the first integer m_0 such that $\sum_{i \neq j} H(j, i, n) < \theta + \epsilon$ for all $n \geq m_0$ with

$$\theta = \sum_{i \neq j} \theta_{ij} \text{ and } \theta_{ij} = (\sigma_i^2 + \sigma_j^2)^{1/2} \text{ and } \epsilon > 0 \text{ arbitrary.}$$

(iii) a_0 is the first integer for which $h(t, n) < \beta$ for all $n \geq a_0$ with $\beta = \delta/(\theta + \epsilon)$.

It follows from the Strong Law of Large Numbers that $M' < \infty$ a.s. subject to (32).

Lemma 4

$$N' \leq M' \text{ a.s.}$$

Proof

If $N' \geq M'$, then $I_{M'-1}$ contains at least $m+1$ indices, say $j_1 < \dots < j_\ell$ (some $\ell = m+1, \dots, k$). From the definition of M' and for any $j_i \in \{1, \dots, k-m\}$ we obtain

$$\begin{aligned} \max_{j \in I_{M'-1}} \bar{X}_j(M') &> \bar{X}_{j_i}(M') + \delta > \bar{X}_{j_i}(M') + h(t, M') \sum_{i \neq j} H(j, i, M') \\ &\geq \bar{X}_{j_i}(M') + h(t, M') H(j_s, j_i, M') \text{ for all } s = 1, \dots, \ell \end{aligned}$$

so that $N' = M'$. Hence $N' \leq M'$ a.s.

Lemma 5

(i) $M'/\log t \rightarrow L$ a.s. as $P^* \rightarrow 1$ where $L = 1/\log(1 + \beta^2)$.

(ii) The set $\{M'/\log t\}$ for $P^* < 1$ is uniformly integrable.

Proof

(i) From the facts that $a_0 \rightarrow \infty$, $h(t, a_0) < \beta$, $h(t, a_0 - 1) \geq \beta$ we obtain

$$a_0/\log t \rightarrow L \text{ as } P^* \rightarrow 1.$$

Since $R_1, \dots, R_{k-m+1}$, R^* do not depend on P^* , it follows that the limit of $M'/\log t$ is also L .

(ii) Let r be the largest integer less than or equal to $n \log t$. Then for $i = 1, \dots, k-m+1$

$$\begin{aligned} &P(R_i/\log t > n) \\ &\leq P(R_i > r) \\ &= P(|\bar{X}_i(m) - \mu_i| \geq \delta \text{ for some } m \geq r) \\ &\leq \sum_{m=r}^{\infty} P(|\bar{X}_i(m) - \mu_i| \geq \delta) \\ &\leq \sum_{m=r}^{\infty} E(\bar{X}_i(m) - \mu_i)^6 / \delta^6 \\ &\leq K \sum_{m=r}^{\infty} m^{-3} \text{ with } K \text{ independent of } P^* \\ &= O(n^{-2}) \text{ uniformly in } P^* \geq P_0^* \text{ (some } P_0^* < 1) \end{aligned}$$

Also $P(R^*/\log t > n)$

$$\begin{aligned}
&\leq P\left(\sum_{i \neq j} H(j, i, m) \geq \theta + \epsilon \text{ for some } m \geq r\right) \\
&\leq \sum_{m=r}^{\infty} P\left(\sum_{i \neq j} H(j, i, m) \geq \theta + \epsilon\right) \\
&= \sum_{m=r}^{\infty} P\left(\sum_{i \neq j} (H(j, i, m) - \theta_{ij}) \geq \epsilon\right) \\
&\leq \sum_{i \neq j} \sum_{m=r}^{\infty} P(H(j, i, m) - \theta_{ij} \geq \gamma) \text{ where } \gamma = \epsilon/k(k-1) \\
&\leq \sum_{i \neq j} \sum_{m=r}^{\infty} P(H^2(j, i, m) - \theta_{ij}^2 \geq \gamma^2) \\
&\leq \sum_{i \neq j} \sum_{m=r}^{\infty} P(|H^2(j, i, m) - \theta_{ij}^2| \geq \gamma^2) \\
&\leq \sum_{i \neq j} \sum_{m=r}^{\infty} E(H^2(j, i, m) - \theta_{ij}^2)^2 / \gamma^4 \\
&\leq K \sum_{m=r}^{\infty} m^{-3} \text{ with } K \text{ independent of } P^* \\
&= O(n^{-2}) \text{ uniformly in } P^* \geq P_0.
\end{aligned}$$

It follows again from the result of Bickel and Yahav (1968) that $R_i/\log t$, $(i=1, \dots, k-m+1)$ and $R^*/\log t$ are uniformly integrable, so that also $M'/\log t$ is uniformly integrable.

If T' denotes the total number of observations at stage N' , then we may now state the following theorem, the proof of which is exactly the same as the proof of theorem 3, except that N , T and M are replaced by N' , T' and M' respectively:

Theorem 4

If $\mu_k > \mu_{k-1}$ and subject to (32) we have

$$\begin{aligned}
\lim_{P^* \rightarrow 1} ET' / (-4 \log(1 - P^*)) &= \sum_{i=1}^{k-m-1} x_{(i)} + (m+1)x_{(k-m)} \\
\lim_{P^* \rightarrow 1} EN' / (-4 \log(1 - P^*)) &= x_{(k-m)}
\end{aligned}$$

where $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(k-1)}$ are the values of $x_1, \dots, x_{k-1}$ arranged in nondecreasing order and

$$x_i = 1 / \log \{1 + (\mu_k - \mu_i)^2 / (\sigma_k^2 + \sigma_i^2)\} \text{ for } i = 1, \dots, k-1.$$

4. **Selection sequences for the case of a known common variance**

We now assume that $\sigma_1^2 = \dots = \sigma_k^2 = \sigma^2$, a known constant. Let

$$g(n, b) = ((b^2 + \log n)/n)^{1/2}$$

where $b = b(k, P^*)$ satisfies the equation

$$(33) \quad 1 - \Phi(b) + b \phi(b) + \phi^2(b)/\Phi(b) = (1 - P^*)/(k - 1)$$

where Φ and ϕ are the cdf and density of the standard normal distribution. It follows that $b \geq 0$ for $k \geq 2$ and $P^* \geq (\pi - 2)/2\pi$, $b \uparrow \infty$ as $P^* \uparrow 1$ and

$$(34) \quad b^2 \sim 2 \log((k-1)/(1 - P^*)) \text{ as } P^* \rightarrow 1.$$

Let, for $n \geq 1$,

$$\Pi_n = \{\pi_s : \bar{X}_s(n) > \bar{X}_1(n) - c_n(P^*), s \neq 1, s \in I_{n-1}\} \cup \{\pi_1\}$$

where $I_0 = \{1, \dots, k\}$, $I_n = \{s : \pi_s \in \Pi_n\}$, $n \geq 1$

and where $c_n(P^*) = -\Delta^* + 2^{1/2} \sigma g(n, b)$ for $\Delta^* > 0$.

That the sequence $\{\Pi_n\}$ is a selection sequence for $\mu_k - \mu_{k-1} \geq \Delta^*$ follows in the same way as theorem 1 was proved and using the following result of Robbins (1970).

Lemma 6.

Let $\{X_n\}$ be a sequence of independent random variables, each having a $N(\mu, \sigma^2)$ distribution. Let $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$. Then

$$P(\bar{X}_n - \mu \geq \sigma g(n, b) \text{ for some } n \geq 1)$$

$$\leq 1 - \Phi(b) + b\phi(b) + \phi^2(b)/\Phi(b) \text{ where } g(n, b) \text{ is defined in (33).}$$

Since the sequence $c_n(P^*)$ is a non-random sequence which decreases with n (n large enough), there is a smallest integer w such that, for $n \geq w$, $c_n(P^*) < 0$, so that I_n contains exactly one population. If x is the (unique) solution of

$$(35) \quad g(x, b) = \Delta^*/2^{1/2} \sigma$$

and v is the largest integer less than x , we have $w = v + 1$. We shall say that w is a *truncation point* since nothing can be gained by computing the selection sequence for $n > w$. The selection sequence thus becomes a finite sequence $\{\Pi_n, 1 \leq n \leq w\}$.

It is interesting to note that, if n^* denotes the sample size of the fixed-sample procedure of Bechhofer (1954) then $w/n^* \rightarrow 1$ as $P^* \rightarrow 1$. In fact $(\Delta^*)^2 w / 2b^2 \sigma^2 \rightarrow 1$ as $P^* \rightarrow 1$ (from (35)) and

$$(36) \quad n^* = \sigma^2 h^2 / (\Delta^*)^2$$

where h is the solution of the equation

$$\int_{-\infty}^{\infty} \Phi^{k-1}(y+h) d\Phi(y) = P^*$$

and $h^2 \sim 4 \log(1 - P^*)^{-1}$ (see Bechhofer, Kiefer and Sobel (1968), p. 185). Now use (34). Thus we have constructed a selection sequence with number of steps asymptotically no more than the number of observation vectors in the standard fixed-sample procedure. Since the selection sequence has the elimination property, it is to be anticipated that sequential procedures based on it will be considerably more efficient than the fixed-sample procedure. That this is indeed the case will be proved shortly.

A sequential procedure based on the selection sequence

As in section 2 we define the stopping variable N to be the first $n \geq 1$ such that $S_n \leq m$ where S_n is the number of observations in Π_n and where $1 \leq m \leq k-1$. Choose Π_N as subset which contains the best population. It follows immediately that Π_N contains at most m populations and that $P(\text{CS}) \geq P^*$ for $\mu_k - \mu_{k-1} \geq \Delta^*$. Denote by T the total number of observations taken by the procedure, then the following theorem may be proved in a way analogous to the proof of theorem 3 (where w now plays the role of M).

Theorem 5

$$\begin{aligned} \lim_{P^* \rightarrow 1} ET / (-2 \log(1 - P^*)) \\ = 2\sigma^2 \left\{ \sum_{i=1}^{k-m-1} (\mu_k - \mu_i + \Delta^*)^{-2} + (m+1)(\mu_k - \mu_{k-m} + \Delta^*)^{-2} \right\} \\ \lim_{P^* \rightarrow 1} EN / (-2 \log(1 - P^*)) = 2\sigma^2 (\mu_k - \mu_{k-m} + \Delta^*)^2 \end{aligned}$$

Asymptotic efficiency

We shall now compare the above procedure with $m = 1$ with several existing proce-

dures. Let T_A and T_B denote the total number of observations taken by two selection procedures A and B respectively. Let P_A and P_B be the probabilities of a correct selection. Then we define the asymptotic efficiency as $P^* + 1$ of procedure A relative to procedure B as

$$\lim_{P^* \rightarrow 1} ET_B/ET_A$$

provided the limit exists and $P_A \geq P^*$, $P_B \geq P^*$ when $\mu_k - \mu_{k-1} \geq \Delta^*$.

From (36) and theorem 5 we see that the asymptotic efficiency of Bechhofer's fixed-sample procedure relative to our procedure is given by

$$\begin{aligned} & (\Delta^*)^2 \left\{ \sum_{i=1}^{k-1} (\mu_k - \mu_i + \Delta^*)^{-2} + (\mu_k - \mu_{k-1} + \Delta^*)^{-2} \right\} / k \\ & \leq (\Delta^*)^2 (k/4(\Delta^*)^2) / k \text{ for } \mu_k - \mu_i \geq \Delta^*, i=1, \dots, k-1 \\ & = 1/4. \end{aligned}$$

The asymptotic efficiency of the procedure of Paulson (1964) relative to our procedure is

$$(37) \quad e(\lambda) = 4(\Delta^* - \lambda) \frac{\left\{ \sum_{i=1}^{k-1} (\mu_k - \mu_i + \Delta^*)^{-2} + (\mu_k - \mu_{k-1} + \Delta^*)^{-2} \right\}}{\left\{ \sum_{i=1}^{k-1} (\mu_k - \mu_i + \lambda)^{-1} + (\mu_k - \mu_{k-1} + \lambda)^{-1} \right\}}$$

where λ is a parameter of Paulson's procedure with $0 \leq \lambda < \Delta^*$. The expression for $e(\lambda)$ follows from a result of Perng (1969) and theorem 5. We now prove that $e(\lambda) \leq 1$ for $0 \leq \lambda < \Delta^*$ and $\mu_k - \mu_{k-1} \geq \Delta^*$. This will clearly be the case if

$$4(\Delta^* - \lambda)(\delta_{ki} + \Delta^*)^{-2} / (\delta_{ki} + \lambda)^{-1} \leq 1, 1 \leq i \leq k-1$$

where $\delta_{ki} = \mu_k - \mu_i$.

But this inequality may be written as

$$4\lambda(\Delta^* - \delta_{ki} - \lambda) \leq (\delta_{ki} - \Delta^*)^2$$

which is satisfied for all $\lambda \geq 0$ since $\Delta^* - \delta_{ki} \leq 0$ for parameter points outside the indifference zone.

From Bechhofer, Kiefer and Sobel (1968, p. 161) and theorem 5 we obtain the following expression for the asymptotic efficiency of their procedure relative to ours:

$$\begin{aligned}
& 4(\Delta^*)^2 \left\{ \sum_{i=1}^{k-1} (\mu_k - \mu_i + \Delta^*)^{-2} + (\mu_k - \mu_{k-1} + \Delta^*)^{-2} \right\} / k \\
& \leq 4(\Delta^*)^2 (k/4(\Delta^*)^2) / k \text{ for } \mu_k - \mu_i \geq \Delta^*, i = 1, \dots, k-1 \\
& = 1.
\end{aligned}$$

Thus our procedure is asymptotically at least as efficient as each of the last two procedures. It may be noted that the last two efficiencies computed may be considerably less than 1. We remark that in Swanepoel and Venter (1975) a procedure is considered which is asymptotically equivalent to our procedure and which is asymptotically optimal as $P^* \rightarrow 1$ within a certain class of procedures.

Remark

So far we have developed asymptotic theory as $P^* \rightarrow 1$. Another possibility which is often considered in this area is to let $\Delta^* \rightarrow 0$. However, although we may prove that for each of the three procedures which was compared with our procedure the total number of observations tends to infinity a.s. for P^* fixed as $\Delta^* \rightarrow 0$, we do not obtain such a result for our procedure. In fact, if we denote by N' the stopping variable N with Δ^* replaced by 0 everywhere in its definition, we obtain

$$\limsup_{\Delta^* \rightarrow 0} N \leq N'$$

where (a) $N' < \infty$ a.s. if $\mu_1 < \dots < \mu_{k-m+1} \leq \dots \leq \mu_k$

(this follows as in lemma 4)

(b) $P(N' < \infty) > 0$ for other configurations.

This property of our procedure is an important one, since in practice one would usually prefer a small value of Δ^* , and for Δ^* small enough and for typical configurations (such as those in (a)) our procedure should take fewer observations than its competitors above.

A selection sequence with no indifference zone assumption

Exactly as was done in section 3 we may obtain a selection sequence without an indifference zone by putting $\Delta^* = 0$ in the definition of the selection sequence defined at the beginning of this section. Since $c_n(P^*) \geq 0$ the sequence is infinite. If we again denote by N' the first stage such that a member of the selection sequence contains at

most m populations and by T' the total number of observations taken up to that stage it is proved as in section 3 that $N' < \infty$ a.s. if $\mu_1 < \dots < \mu_{k-m+1} \leq \dots \leq \mu_k$ and in this case asymptotic expressions as $P^* \rightarrow 1$ may be derived for EN' and ET' in the same way as was done in section 3. The expressions are those given in theorem 5 with $\Delta^* = 0$.

We again stress the importance of this sequence : no indifference zone assumption is required and the sampling may be stopped at an arbitrary point. In typical cases the sequence shrinks to one population only.

Tables

For practical applications of the procedures, we give the following table I with values of the constant b , defined at the beginning of section 4. A table with values of the constant a defined at the beginning of section 2 is contained in Swanepoel and Geertsema (1973).

TABLE I

P^* k	0,65	0,70	0,75	0,80	0,85	0,90	0,95	0,99
2	1,306	1,441	1,586	1,746	1,931	2,163	2,503	3,137
3	1,835	1,931	2,039	2,163	2,311	2,503	2,797	3,368
4	2,078	2,163	2,258	2,369	2,503	2,679	2,953	3,496
5	2,233	2,311	2,400	2,503	2,629	2,797	3,058	3,583
6	2,345	2,419	2,503	2,602	2,723	2,884	3,137	3,649
7	2,432	2,503	2,584	2,679	2,797	2,953	3,200	3,702
8	2,503	2,572	2,650	2,743	2,857	3,010	3,252	3,746
9	2,563	2,629	2,706	2,797	2,908	3,058	3,296	3,784
10	2,614	2,679	2,754	2,843	2,953	3,100	3,334	3,817
11	2,659	2,723	2,797	2,884	2,992	3,137	3,368	3,847
12	2,699	2,762	2,834	2,920	3,027	3,170	3,399	3,873
13	2,734	2,797	2,868	2,953	3,058	3,200	3,426	3,897
14	2,767	2,828	2,899	2,982	3,086	3,227	3,451	3,918
15	2,797	2,857	2,927	3,010	3,113	3,252	3,474	3,938

Monte Carlo results

Limited Monte Carlo studies were carried out to compare the procedure which was introduced in section 4 in the case $m = 1$ (designated procedure X) with other procedures and to see whether the asymptotic results of theorem 5 are good approximations to

actual sample sizes.

Random samples from k normal populations were generated for $k = 5, 10$ and 15 and both procedures X and the procedure of Paulson (procedure P) were applied for various values of Δ^* and P^* to the same samples. The latter procedure was applied only with $\lambda = 0$ (see Ramberg (1966) p. 65, 86). Let $\mu_k - \mu_{k-1} = \delta$ and $\mu_i - \mu_{i-1} = \frac{1}{2}$, $i = 2, \dots, k-1$, then table II gives the result for the case $\delta = \frac{1}{2}$ (Studies with $\delta = 0,3$ were performed also but are not tabulated since the conclusions are similar.) All the results in table II are for $\sigma^2 = 1$. The following are given in table II for procedures X and P :

- (i) the asymptotic expected total number of observations (AT), computed from theorem 5 and Perng (1969) theorem 5 (the value for $\lambda = 0$ may be obtained by substitution in Perng's formula, but that this is valid follows by using methods similar to those in the present article)
- (ii) the estimated expected total number of observations (MT) obtained from the Monte Carlo studies
- (iii) the standard error (SE) of the MT values.

For all Δ^* in the case $k = 15$ and also for the cases $k = 5, 10$ with $\Delta^* = 0$, the Monte Carlo results are based on 300 repetitions and in all other cases on 200 repetitions. The sample sizes for the Bechhofer procedure were also computed for the case $k = 5$ and are tabled under B.

The MT-results for $P^* = 0,95$ for $k = 5, 10$ and 15 appear in fig. 1-3 as functions of Δ^* . In each case the point of intersection of the X and P curves is indicated.

The following may be concluded:

- (i) For fixed P^* and k the MT values of the X -procedure are considerably smaller than the corresponding values of the P -procedure for Δ^* small, while for other values of Δ^* the differences are relatively small. Also note the behaviour of the X -procedure when $\Delta^* = 0$.
- (ii) For fixed k and Δ^* the efficiency of the X -procedure with respect to the P -procedure based on the MT-values increases with P^* , in correspondence with the asymptotic results. Note also the correspondence between the MT and AT-values.
- (iii) For fixed P^* and Δ^* the efficiency of the X -procedure with respect to the P -procedure based on the MT-values increases as k increases. This is in agreement with the fact that

$$e(0) \sim C/\log k \quad \text{as } k \rightarrow \infty$$

where C is a constant depending only on Δ^* , and where $e(\lambda)$ is defined in (37).

TABLE II

P*	Δ^*	k=5							k=10							k=15						
		X			P			B	X			P				X			P			
		AT	MT	SE	AT	MT	SE		AT	MT	SE	AT	MT	SE		AT	MT	SE	AT	MT	SE	
0,90	0	214,3	212,9	7,1	—	—	—	—	257,5	263,8	7,8	—	—	—	—	279,4	295,9	7,9	—	—	—	—
	0,05	180,7	179,9	7,2	455,0	459,8	5,5	13516,9	218,6	226,1	7,2	689,2	691,0	6,3	237,7	247,3	6,3	840,4	838,1	5,3	—	—
	0,1	154,7	154,0	6,1	227,5	231,6	4,1	3379,2	188,5	194,4	6,4	344,6	356,2	4,6	205,4	213,3	5,6	420,2	435,8	3,7	—	—
	0,2	117,8	112,0	4,5	113,7	123,1	2,9	844,8	145,7	146,9	5,0	172,3	181,1	3,1	159,3	162,4	4,2	210,1	221,4	2,6	—	—
	0,3	93,2	88,7	3,5	75,8	80,0	2,0	375,5	116,6	107,8	3,6	114,9	119,8	2,4	128,3	126,6	3,1	140,1	151,5	2,3	—	—
	0,4	75,8	69,9	2,6	56,9	62,8	1,8	211,2	96,2	92,3	2,8	86,1	91,0	1,8	106,3	108,9	2,5	105,0	116,5	2,0	—	—
	0,5	63,1	55,1	1,9	45,5	49,6	1,4	135,2	81,1	78,2	2,4	68,9	76,3	1,8	90,1	91,3	2,0	84,0	92,6	1,5	—	—
0,95	0	241,2	251,3	7,5	—	—	—	—	285,7	299,7	7,9	—	—	—	—	308,0	327,4	8,4	—	—	—	—
	0,05	203,3	208,2	8,0	540,4	538,6	5,4	18668,5	242,5	250,5	8,4	795,3	820,5	6,4	262,0	285,6	6,7	958,3	987,7	6,2	—	—
	0,1	174,1	176,9	6,8	270,2	277,3	4,6	4667,1	209,1	218,6	6,8	397,7	408,5	4,9	226,4	237,5	5,9	479,1	489,4	4,2	—	—
	0,2	132,6	133,2	4,7	135,1	141,8	2,9	1166,8	161,4	157,2	5,2	198,8	209,5	3,7	175,6	181,2	4,1	239,6	246,1	2,7	—	—
	0,3	104,8	99,4	3,5	90,1	95,3	2,1	518,6	129,4	139,7	4,1	132,6	139,9	2,5	141,4	148,3	3,2	159,7	166,7	2,2	—	—
	0,4	85,3	83,0	3,0	67,6	70,0	1,7	291,7	106,7	106,4	2,8	99,4	105,1	2,1	117,2	126,0	2,8	119,8	132,7	2,1	—	—
	0,5	71,0	68,6	2,3	54,0	55,9	1,6	186,7	90,0	86,1	2,5	79,5	88,3	1,8	99,3	98,8	2,2	95,8	106,0	1,7	—	—
0,99	0	303,6	334,3	8,4	—	—	—	—	351,1	394,3	8,8	—	—	—	—	374,3	389,3	8,9	—	—	—	—
	0,05	256,0	272,0	8,7	738,9	756,6	7,1	30726,5	298,0	315,8	9,0	1041,8	1063,0	8,1	318,4	346,3	7,7	1232,0	1258,8	5,9	—	—
	0,1	219,2	237,1	7,1	369,5	375,2	4,8	7681,6	257,0	285,7	8,5	520,9	531,9	5,4	275,2	295,6	6,4	616,0	631,6	5,4	—	—
	0,2	166,9	164,9	5,1	184,7	189,3	3,7	1920,4	198,3	209,7	6,2	260,5	270,0	3,9	213,4	220,9	4,7	308,0	320,6	3,2	—	—
	0,3	132,0	132,6	4,0	123,2	129,3	2,8	853,5	159,0	170,8	4,3	173,6	185,7	3,1	171,9	179,6	3,7	205,3	217,9	2,6	—	—
	0,4	107,4	109,2	3,0	92,4	103,0	2,4	480,1	131,1	138,2	3,2	130,2	137,8	2,7	142,4	152,3	3,1	154,0	165,6	2,4	—	—
	0,5	89,4	90,8	2,8	73,9	80,3	2,2	307,3	110,6	110,1	2,9	104,2	109,6	2,2	120,7	123,1	2,3	123,2	133,1	2,0	—	—

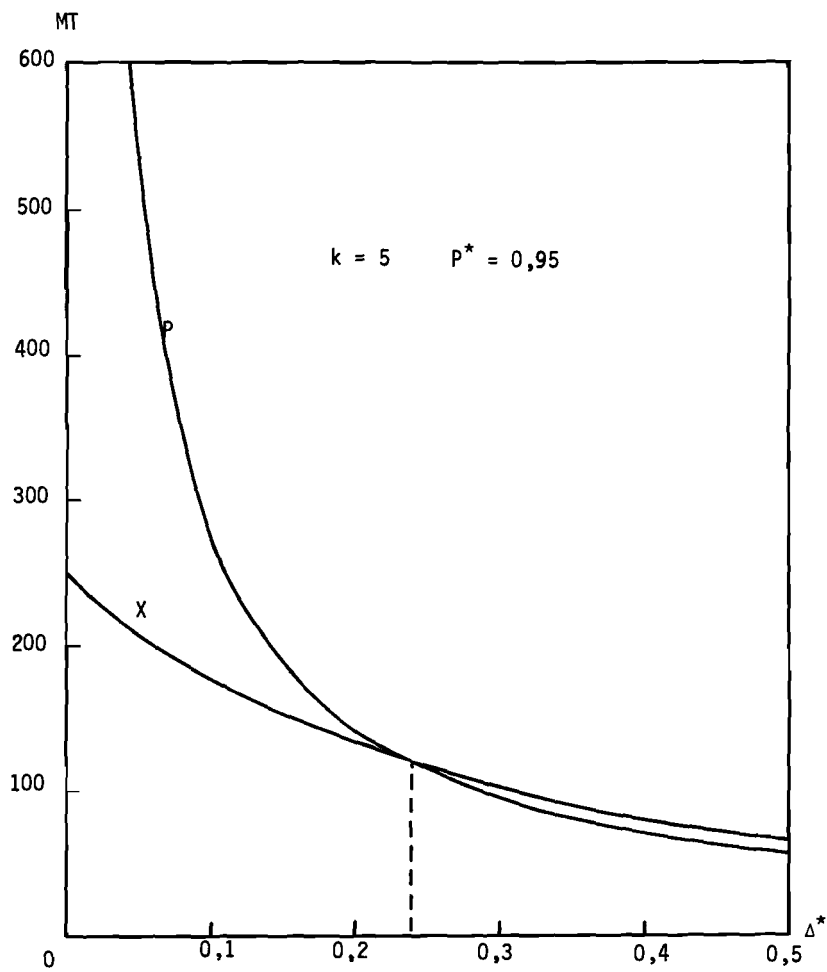


FIGURE 1

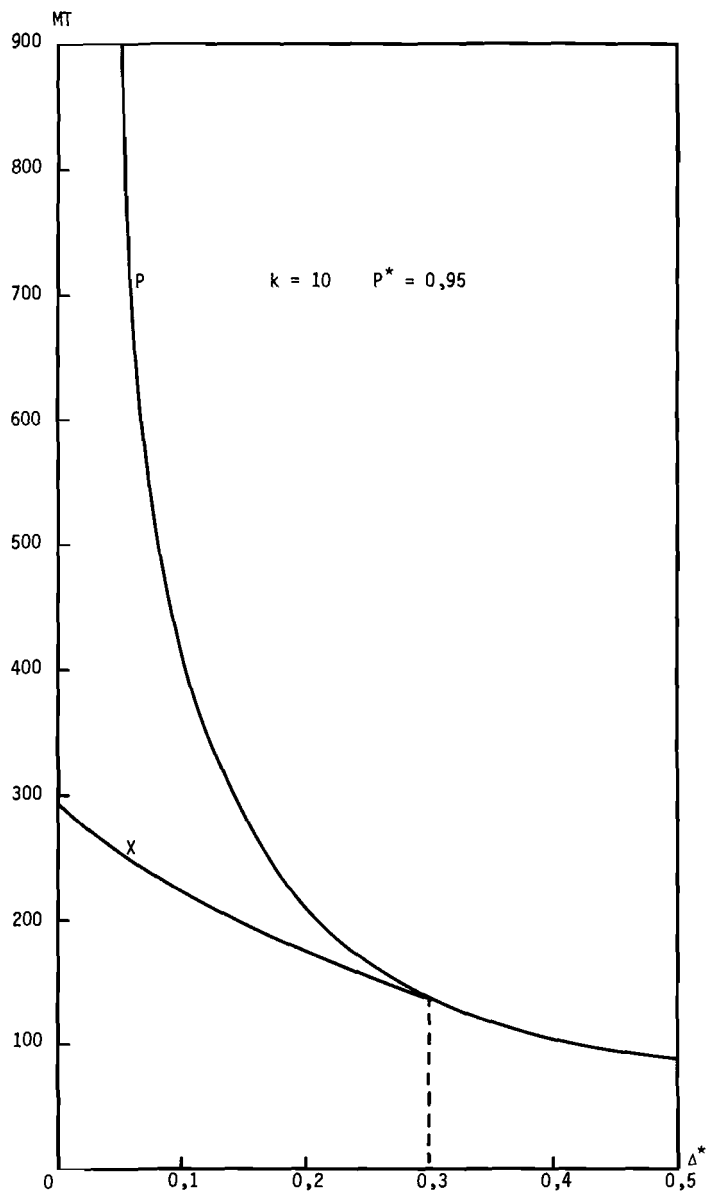


FIGURE 2

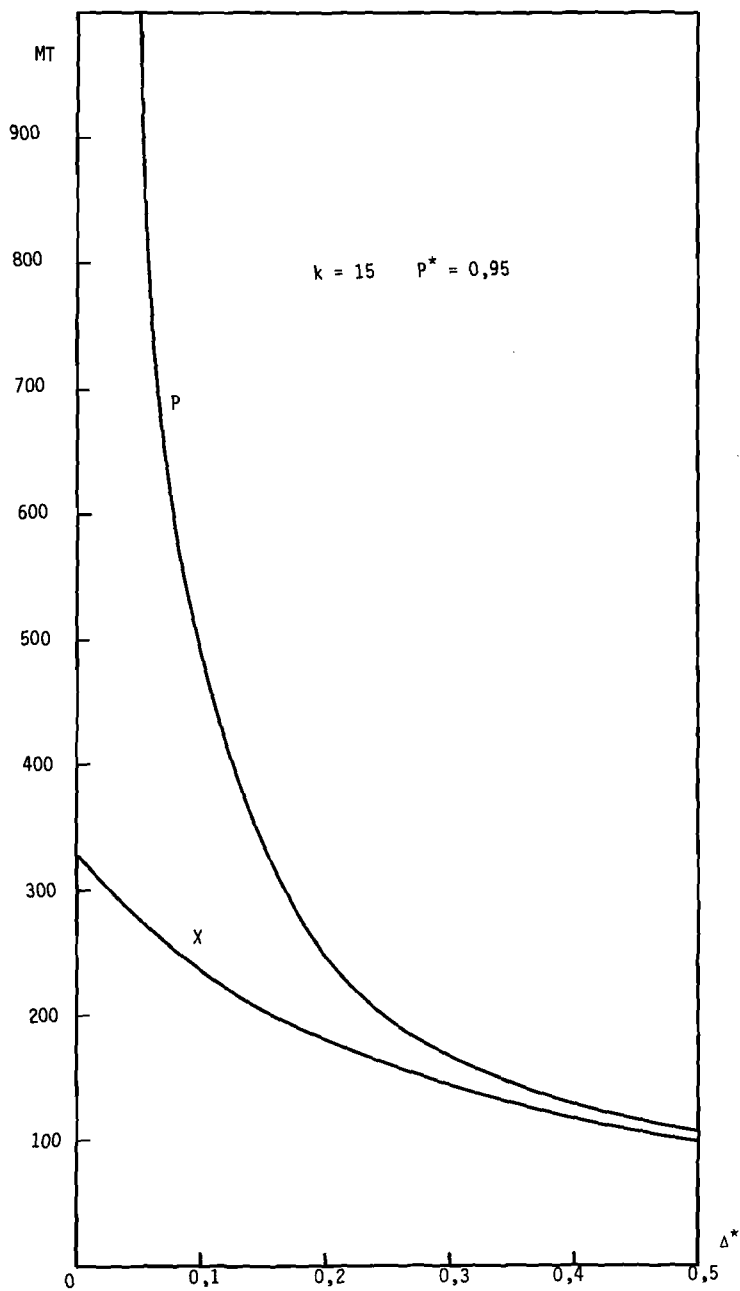


FIGURE 3

In order to compare the asymptotic results for the procedures in sections 2 and 3 (theorems 3 and 4) with actual sample sizes, table III was constructed based on 400 Monte Carlo runs. A few values of P^* and $k = 5$ were chosen and $\delta = 0,5$. The following configurations of variances were considered:

I $\sigma_1^2 = 0,2; \sigma_2^2 = 0,4; \sigma_3^2 = 0,6; \sigma_4^2 = 0,8; \sigma_5^2 = 1,0$

II $\sigma_1^2 = 0,2; \sigma_2^2 = 0,4; \sigma_3^2 = 0,6; \sigma_4^2 = 0,4; \sigma_5^2 = 0,2$

III $\sigma_1^2 = 1,0; \sigma_2^2 = 0,8; \sigma_3^2 = 0,6; \sigma_4^2 = 0,4; \sigma_5^2 = 0,2$

TABLE III

P^*	0,85			0,90			0,95		
Configuration	I			II			III		
Δ^*	AT	MT	SE	AT	MT	SE	AT	MT	SE
0,0	339,4	380,8	16,3	155,3	161,8	5,1	188,0	207,4	5,4
0,1	254,2	279,1	11,5	122,7	130,5	3,5	149,8	160,3	3,8
0,3	166,3	160,8	5,4	88,0	85,1	1,8	108,8	108,0	2,0
0,5	123,2	111,9	3,3	70,2	66,0	1,1	87,4	83,5	1,3

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